

Regime Detection using Sliced Wasserstein k-means clustering with Monte Carlo

Oscar Peyron

ADIA Lab - University College London

February 2026

1 Abstract

This thesis investigates the topic of Regime Detection using an interpretable Machine Learning algorithm. In particular, the focus of this thesis is to produce a tractable algorithm for detecting regimes in financial markets based on multi-dimensional time series. This thesis, provides an enhanced algorithm to cluster the multi-dimensional returns distributions. Depending, on the the characteristics of the distributions, the algorithm can infer which regime the market is in without pre-specified assumptions on the returns distribution. Specifically, it uses the Wasserstein distance as a metric to compare different regimes. In higher-dimensional settings, one can extend this concept using the Sliced Wasserstein distance and use an algorithm based on this metric to cluster different market regimes (Sliced Wasserstein k-means clustering algorithm). The following thesis is structured as follows believed to be the most explanatory and logical way to present the research.

2 Theoretical Background

This chapter will focus on the theoretical framework to understand the Wasserstein distance and its extension to the Sliced Wasserstein distance. It will also provide the mathematics behind the implied probability derived from the Wasserstein distance to produce predictive signals for trading strategies. Finally, we will provide some background on the statistical tests we used to compare the performance of the different trading strategies (i.e Ledoit-Wolf test on Sharpe ratios).

2.1 Theoretical Framework for Wasserstein Distance and Wasserstein Barycenters

In this chapter we consider $\mathcal{X} = \mathbb{R}^d$ with d the dimension of the returns vector.

Definition: Data Streams A set $\mathcal{S}$ of streams of data over $\mathcal{X}$ is defined as:

$$\mathcal{S}(\mathcal{X}) = \{(X_1, X_2, \dots, X_N) | X_i \in \mathcal{X}, N \in \mathbb{N}\}.$$

For instance in this research we consider the following:

$$S = (S_1, S_2, \dots, S_N) \in \mathcal{S}(\mathbb{R}^d) \quad (1)$$

where S_i might be the the price vector of the different assets at time i . [2]

Definition: Lifting Transformation A lifting transformation l is a map that maps the set of streams $\mathcal{S}(\mathcal{X})$ into the set of streams of streams:

$$l: \begin{cases} \mathcal{S}(\mathcal{X}) \rightarrow \mathcal{S}(\mathcal{S}(\mathcal{X})) \\ S \mapsto l(S) = (l_1, l_2, \dots, l_M) \end{cases} \quad (2)$$

where $M > 1$. [2]

The example from [1] is the sliding window transformation:

$$l^m(\mathbf{x}) = (x_{1+h_2(m-1)}, x_{2+h_2(m-1)}, \dots, x_{h_1+h_2(m-1)}) \text{ for } m = 1, 2, \dots, M \quad (3)$$

with $M := \left\lfloor \frac{N-h_1-h_2}{h_2} \right\rfloor$ and $h_1, h_2 \in \mathbb{N}$ the window size and the step size respectively.

Definition: Discrete uniform distribution Given a stream generated by the lifting process $l(S) = (l^m(\mathbf{x}))_i, : i = 1, 2, \dots, h_1)$, we can define the discrete uniform distribution μ^S as follows:

$$\mu_m = \frac{1}{h_1} \sum_{i=1}^{h_1} \delta_{l^m(\mathbf{x})_i} \quad (4)$$

where we have that δ_x is the Dirac delta at point x . [4] This definition allows us to introduce the set of empirical distributions $\mathcal{K} = \{\mu_j\}_{1 \leq j \leq M}$ where $\mu_j \in \mathcal{P}(\mathbb{R}^d)$ with $\mathcal{P}(\mathbb{R}^d)$ the set of probability measures on $\mathbb{R}^d$ with finite p -th moment. [4].

Definition: Wasserstein distance $\mathcal{W}_p$ Let μ and ν be two probability measures on $\mathbb{R}^d$ with finite p -th moment defined as the following infimum over the joint distributions (X, Y) of d -dimensional random vectors X and Y . The Wasserstein distance of order p between μ and ν is defined as:

$$\mathcal{W}_p(\mu, \nu) = \inf_{(X, Y) \sim \Gamma(\mu, \nu)} (\mathbb{E} [\|X - Y\|^p])^{1/p} \quad (5)$$

where $\Gamma(\mu, \nu)$ is the set of all couplings of μ and ν , i.e., the set of all probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν . [4]

Definition: Wasserstein barycentre $\bar{\mu}$ Given a family probability measures $\mathcal{K}$ in $\mathbb{R}^d$ with finite p -th moment, the Wasserstein barycentre $\bar{\mu}$ is defined as the probability measure that minimizes the sum of squared Wasserstein distances to the given measures:

$$\bar{\mu} = \arg \min_{\nu \in \mathcal{P}(\mathbb{R}^d)} \sum_{\mu_m \in \mathcal{K}} \mathcal{W}_p(\nu, \mu_m) \quad (6)$$

where $\mathcal{P}(\mathbb{R}^d)$ is the set of probability measures on $\mathbb{R}^d$ with finite p -th moment. [4]

Projected Empirical Measure Let $\theta \in \mathbb{S}^{d-1}$ with $\mathbb{S}^{d-1}$ the unit sphere in d dimensions. The projected empirical measure $\mu'(\theta) \in \mathcal{P}(\mathbb{R})$ is defined as:

$$\mu'(\theta) = \frac{1}{N} \sum_i \delta_{\langle x_i, \theta \rangle} \quad (7)$$

with x_i the atoms of the original empirical measure $\mu \in \mathcal{P}(\mathbb{R}^d)$ and $\langle \cdot, \cdot \rangle$ the standard inner product in $\mathbb{R}^d$. [4]

Definition: The sliced Wasserstein distance $\bar{\mathcal{W}}_p$ Given two distributions $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$ with $d > 1$, the sliced Wasserstein distance is defined as follows:

$$\bar{\mathcal{W}}_p(\mu, \nu) = \int_{\mathbb{S}^{d-1}} \mathcal{W}_p(\mu'(\theta), \nu'(\theta)) d\theta \quad (8)$$

with $\mathcal{W}_p(\mu'(\theta), \nu'(\theta))$ the Wasserstein distance in $d=1$ between the projected empirical measures $\mu'(\theta)$ and $\nu'(\theta)$ on the direction θ . [4]

Definition: The sliced Wasserstein Barycenter

2.2 Theoretical Framework for Sliced Wasserstein (SW) with UnifOrtho Projections

This section will provide insights on the practical implementation of the Sliced Wasserstein k-mean clusterings algorithms and limitations to its practical implementations in the literature. Notably, the low generalizable nature of using pre-defined projection vectors θ^l for the Sliced Wasserstein can be solved by generating them using the UnifOrtho method [5]. It provides the following advantages compared to standard Monte Carlo methods.

- **Variance reduction via Negative Negative Dependence**
- **Superior performance in high dimensions** : The UnifOrtho is therefore perfectly adapted to working with multi-dimensional time series for the SW k-means clustering algorithm.
- **Clustering Stability**

Some useful theoretical results on variance reduction and computational cost on the UnifOrtho method will therefore be provided to support the choice of this method for the future sections.

The following pseudocode is based on the work by [5] on Repulsive Monte Carlo Methods and [4] on Sliced Wasserstein k-means clustering for regime detection.

Algorithm 1: sWk-means clustering with UnifOrtho Projections

Input: Data stream $S \in \mathcal{S}(\mathbb{R}^d)$;

number of clusters K ;

number of projections L ;

tolerance ϵ for convergence

Output: K cluster assignments and corresponding 1d barycenters

▷ **Preprocessing**

1 Calculate $l(r^S)$ given $S \in \mathcal{S}(\mathbb{R}^d)$ and define family of empirical distributions $\mathcal{K} = \{\mu_j\}_{1 \leq j \leq M}$;

▷ **Step 1: UnifOrtho Projection Generation**

2 $k \leftarrow \lceil L/d \rceil$;

▷ number of orthogonal bases needed

3 $\Theta \leftarrow \emptyset$;

4 **for** $i = 1$ **to** k **do**

5 Sample a random matrix $Z_i \in \mathbb{R}^{d \times d}$ with i.i.d. $\mathcal{N}(0, 1)$ entries;

6 Compute the QR decomposition: $Z_i = Q_i R_i$;

7 Let Λ_i be a diagonal matrix where $\Lambda_{i,jj} = \text{sgn}(R_{i,jj})$;

8 $U_i \leftarrow Q_i \Lambda_i$;

▷ U_i is Haar-distributed on $O(d)$

9 Add the d column vectors of U_i to Θ ;

10 $\{\theta^1, \dots, \theta^L\} \leftarrow \Theta[1 : L]$;

▷ select exactly L directions

▷ **Step 2: Projection & K-Means Iteration**

11 Calculate projected distributions $\{\mu'_j(\theta^l)\}_{1 \leq j \leq M}$ for all $l = 1 \dots L$;

12 Initialize cluster assignments $C_1, \dots, C_K$;

13 **while** *convergence criterion* $> \epsilon$ **do**

14 **for** $k = 1$ **to** K **do**

15 Compute the 1d Wasserstein barycenter $\bar{\mu}_k(\theta^l)$ for each projection l using empirical distributions assigned to C_k ;

16 **for** $j = 1$ **to** M **do**

17 Assign μ_j to the cluster C_{k^*} where $k^* = \arg \min_k \sum_{l=1}^L W_p^p(\mu'_j(\theta^l), \bar{\mu}_k(\theta^l))$;

18 **return** $C_1, \dots, C_K$ and barycenters $\{\bar{\mu}_k(\theta^l)\}$

2.3 Theoretical Framework for Implied Probability Regime Detection Strategy

Theoretical Background We use implied probability measure to compute whether the market is going to become bear or bull.

Likelihood Function It uses SW-distance to every centroid to every centroid and then convert the likelihood via a softmax over the negative distances. We introduce a temperature parameter τ

$$\mathcal{L}(\text{regime } k \mid x_m) = \frac{e^{-d(x_m, c_k)/\tau}}{\sum_j e^{-d(x_m, c_j)/\tau}}$$

Empirical Transition Matrix Markov Prior A market in regime i is highly likely to stay in regime i tomorrow. We model this as a first-order Markov Chain using the historical labels. Probability of switching from i to j is:

$$\pi_{i,j} = P(Z_m = j \mid Z_{m-1} = i) = \frac{\text{Count}(i \rightarrow j)}{\sum_{x=1}^K \text{Count}(i \rightarrow x)}$$

If the market was z_{current} our prior belief that state M is going to be at regime k is :

$$\mathbb{P}_{\text{prior}}(k) = \pi_{z_{\text{current}}, k}$$

Then we simply use Bayes rule to derive the following :

$$\mathbb{P}_{Bayes}(k) = \frac{\mathbb{P}_{prior}(k) \cdot \mathcal{L}(k | X_m)}{\sum_{j=1}^K \mathbb{P}_{prior}(j) \cdot \mathcal{L}(j | X_m)}$$

Switch Probability Finally the switch probability does the following :

$$\mathbb{P}_{switch} = 1 - \mathbb{P}_{Bayes}(z_{current})$$

Use Gradient One may also want to take into account the fact that one is moving away or getting closer to a certain regime.

We consider a lookback window of size ω , and one computes the linear trend slope of the Wasserstein distance to each centroid k . Let $t \in \{1, 2, \dots, \omega\}$ We get using OLS, β_k :

$$\beta_k = \frac{\text{Covariance}(t, \mathcal{W}_2(X_{M-w+t}, C_k))}{\text{Variance}(t)}$$

- if $\beta_k < 0$ the distance to regime k is shrinking (we are getting close to regime k)
- if $\beta_k > 0$ the distance to regime k is growing (we are drifting away from regime k)

We apply softmax to get a probability:

$$P_{Grad}(k) = \frac{\exp\left(-\frac{\beta_k}{\tau}\right)}{\sum_{j=1}^K \exp\left(-\frac{\beta_j}{\tau}\right)}$$

The final switch probability using gradient is therefore:

$$\mathbb{P}_{switch} = 1 - [(1 - \alpha)\mathbb{P}_{Bayes}(z_{current}) + \alpha\mathbb{P}_{Grad}(z_{current})]$$

with $\alpha \in [0 : 1]$ a weight.

2.4 Theoretical Framework for the Ledoit-Wolf Sharpe ratio test

Need to read the paper [3].

3 Design methodology for Monte Carlo for SW - Synthetic Data Testing

The focus of this chapter will be on testing the algorithm on syntethic data to understand the behaviour of the algorithm and its performance in different settings. The algorithm will be tested on 3 dimensional data followed by 4, 5 dimensional. Performance between original and UniOrtho method will be compared and tests will span from data with multiple correlation factors and with diffrent target number of clusters. Finally, in terms of time-horizon, the study will focus on the equivalent of 20 years synthetic data with 7 observations a day (252 trading days a year) to be as close as possible to real world data.

Different performance metrics will be used for the regime detection :

- Total Accuracy (TA)
- Mean centroid-centroid distance (MCCD)
- Point-to-centroid distance (PCD)

The synthetic data will be generated similarly to previous papers [2] and [4] using Geometric Brownian Motion (GBM) paths $S_t/S_{t-1} = \exp(r_t^S)$ with $S_t = (S_t^1, \dots, S_t^d)$ and $r_t^S = (r_t^{S^{(1)}}, \dots, r_t^{S^{(d)}})$ where :

$$r_t^{S^{(i)}} \sim \mathcal{N}((\mu - \sigma^2/2)1dt, \Sigma dt) \quad (9)$$

. with

$$\Theta = (\mu, \sigma^2) \quad (10)$$

And the covariance matrix Σ is defined as follows :

$$\Sigma_{ij} = \sigma^2 (\delta_{ij} + (1 - \delta_{ij})\rho) \quad (11)$$

The δ_{ij} is the Kronecker delta, σ^2 is the variance of the returns and ρ is the correlation between the different assets.

Assuming, we want to cluster two different regimes (K=2), we define, as in previous papers [4], Bull and Bear regimes with the following parametrization :

- $\Theta_{\text{bull}} = (0.02, 0.2)$
- $\Theta_{\text{bear}} = (-0.02, 0.3)$

For K = 3, similarly to [4] we will consider the *moon-shape structure* generated using `datasets.make_moons` function from the `sklearn` library.

This is a summary similary to the one provided in [4] for the different market settings on which we are going to test the sWk-means clustering algorithm with UnifOrtho projections :

Table 1: Summary of synthetic dataset parameters with $d \in \{3, 4, 5\}$

Type	Regime Bull	Regime Bear	Regime III (Bear/Moon)
Type A (K=2)	$\Theta_{\text{bull}}; \rho = +1/2$	$\Theta_{\text{bear}}; \rho = +1/2$	N/A
Type B (K=2)	$\Theta_{\text{bull}}; \rho = +1/2$	$\Theta_{\text{bull}}; \rho = -1/2$	N/A
Type C (K=3)	$\Theta_{\text{bull}}; \rho = +1/2$	$\Theta_{\text{bear}}; \rho = +1/2$	$\Theta_{\text{bear}}; \rho = -1/2$
Type D (K=3)	$\Theta_{\text{bull}}; \rho = +1/2$	$\Theta_{\text{bear}}; \rho = +1/2$	$\Theta_{\text{moon}}; \rho = +1/2$

4 Application of sliced-SW k-means clustering with UnifOrtho projections on real world data sets

Finally, the chapter will focus on the application of the novel algorithm on real world data sets for different types of financial markets. As above, the time span, wil focus on 15 to 20 years of data. The target data sets include the following :

- FX Spot Series Data (4D)
EURUSD, AUDUSD, USDJPY, USDCHF, USDCAD (Major Currency Cluster) (15 min data)
- Equity Series Data (3D)
S&P 500, NASDAQ, Dow Jones (US Equity Cluster) (daily data)
CAC 40, DAX40, FTSE 100 (European Equity Cluster) (daily data)
Nikkei 225, Hang Seng, Kospi 200 (Asian Equity Cluster) (daily data)
- Commodity Series Data (4D and 5D)
WTI Crude Oil, Brent Crude Oil, Natural Gas, Heating Oil.
Copper Aluminum, Zinc, Nickel, Lead.

4.1 Performance analysis of regime detection using trading strategies

- Long/Short strategy based on majority regime detection signals
- Long/Short strategy based on Implied probability signals (using the distance to the centroids as a proxy for the confidence of the regime detection)
- Ensemble Methods: combining the above two strategies using a weighted average of the signals or an adaptative weighting scheme to generate the final trading signal.

Tested trading strategies on FX and Equity data sets. Performance is compared to a long only and short-only approach. Moreover, for equities, we look at the performance for different clusters (US, Europe, Asia) to see if the performance of the regime detection is better in some markets than others. We also test, trading algorithm that use american regime detection to trade in Asian and European markets to see if the regime detection is more generalizable in some markets than others.

4.1.1 Standard SW k-means trading strategy

Pseudo code for this strategy is provided here :

Algorithm 2: LONG_STRAT_UNIFORTHO — Binary Regime Strategy

Input: Initial capital C_0 ;;
number of simulations N_S ;;
price matrix $S \in \mathbb{R}^{T \times d}$;;
trimming parameters h_1, h_2 ;;
window size w ;;
number of clusters K ;;
risk metric m ;;
majority look-back ℓ
Output: Portfolio value path V

```

1  $R \leftarrow \text{PctReturns}(S)$  ; ▷  $R_t = S_t/S_{t-1} - 1$ 
2  $\theta \leftarrow \mathbf{1}_d / \sigma_d$  ; ▷ inverse vol weighted returns
3  $r^{(\text{pf})} \leftarrow R\theta$  ; ▷ portfolio return series
4  $V \leftarrow [C_0]$ ;
5  $n \leftarrow \lfloor T/w \rfloor$ ;
6 for  $i = 0$  to  $n - 2$  do
7    $s \leftarrow i \cdot w, \quad e \leftarrow (i + 1) \cdot w$ ;
8    $W \leftarrow S[s : e]$  ; ▷ current analysis window
9   ▷ Step 1 -- Wasserstein-based regime detection
10   $\hat{F}, \boldsymbol{\mu}, \boldsymbol{\lambda} \leftarrow \text{MAXMCCD\_UNIFORTHO}(N_S, W, K, L, \varepsilon, h_1, h_2, m)$ ;
11  ▷ Step 2 -- Majority-vote regime classification
12  if  $\ell > |\boldsymbol{\lambda}|$  then
13    |  $\hat{k} \leftarrow \arg \max_k \text{count}(\boldsymbol{\lambda} = k)$ ;
14  else
15    |  $\hat{k} \leftarrow \arg \max_k \text{count}(\boldsymbol{\lambda}_{[-\ell:]} = k)$ ;
16  ▷ Step 3 -- Generate binary signal
17  if  $\hat{k} = 1$  then ▷ Bullish  $\Rightarrow$  Long
18    |  $\sigma \leftarrow +1$  ;
19  else ▷ Bearish  $\Rightarrow$  Short
20    |  $\sigma \leftarrow -1$  ;
21  ▷ Step 4 -- Compound portfolio over next window
22  for each  $r_t \in r^{(\text{pf})}[e : e + w]$  do
23    |  $V.\text{append}(V[-1] \times (1 + \sigma r_t))$ ;
24 return  $V$ 

```

4.1.2 Implied Probability SW k-means trading strategy

Algorithm 3: LONG_STRAT_IMPLIED — Implied-Probability Regime Strategy

Input: Initial capital C_0 ;;
simulations N_S ;;
price matrix $S \in \mathbb{R}^{T \times d}$;;
trimming h_1, h_2 ;;
window w ;;
clusters K ;;
metric m ;;
signal type $\tau \in \{\text{continuous}, \text{hysteresis}, \text{conviction}\}$;;
entry threshold α_e ;;
hold threshold α_h ;;
look-back ℓ ;;
gradient flag and weight $(\mathbf{g}, \omega_g)$
Output: Portfolio value path V ;;
signal history Σ

```

1  $R \leftarrow \text{PctReturns}(S), \quad \theta \leftarrow \mathbf{1}_d \cdot 1/\sigma_d \quad r^{(\text{pf})} \leftarrow R\theta;$ 
2  $V \leftarrow [C_0], \quad \Sigma \leftarrow [], \quad n \leftarrow \lfloor T/w \rfloor;$ 
3 for  $i = 0$  to  $n - 2$  do
4    $s \leftarrow i \cdot w, \quad e \leftarrow (i + 1) \cdot w, \quad W \leftarrow S[s : e];$ 
5    $\hat{F}, \boldsymbol{\mu}, \boldsymbol{\lambda} \leftarrow \text{MAXMCCD\_UNIFORTHO}(N_S, W, K, L, \varepsilon, h_1, h_2, m);$ 
6    $\mathbf{P}, p_{\text{switch}}, \mathbf{T}, \boldsymbol{\pi} \leftarrow \text{COMPUTEIMPLIEDPROBA}(\hat{F}, \boldsymbol{\mu}, \boldsymbol{\lambda}, \ell, \mathbf{g}, \omega_g);$ 
7    $\hat{k} \leftarrow \arg \max_k \text{count}(\boldsymbol{\lambda}_{[-h_2:]} = k);$ 
8   if  $\tau = \text{continuous}$  then
9      $\sigma \leftarrow \pi_{\text{bull}} - \pi_{\text{bear}};$ 
10    if  $|\sigma| < \delta$  then
11       $\sigma \leftarrow 0$ 
12    ; ▷ dead zone
13  else if  $\tau = \text{hysteresis}$  then
14     $\sigma_{\text{prev}} \leftarrow \Sigma[-1]$  or  $0$ ;
15    if  $\hat{k} = \text{Bull}$  then
16      if  $p_{\text{switch}} \geq \alpha_e$  and  $\sigma_{\text{prev}} \geq 0$  then  $\sigma \leftarrow -1$ ;
17      else if  $p_{\text{switch}} < \alpha_h$  and  $\sigma_{\text{prev}} < 0$  then  $\sigma \leftarrow +1$ ;
18      else  $\sigma \leftarrow 0$ ;
19    else ▷  $\hat{k} = \text{Bear}$ 
20      if  $p_{\text{switch}} \geq \alpha_e$  and  $\sigma_{\text{prev}} \leq 0$  then  $\sigma \leftarrow +1$ ;
21      else if  $p_{\text{switch}} < \alpha_h$  and  $\sigma_{\text{prev}} > 0$  then  $\sigma \leftarrow -1$ ;
22      else  $\sigma \leftarrow 0$ ;
23  else if  $\tau = \text{conviction}$  then
24     $d_k \leftarrow \begin{cases} +1 & \hat{k} = \text{Bull} \\ -1 & \hat{k} = \text{Bear} \end{cases};$ 
25     $c \leftarrow 1 - 1.5 p_{\text{switch}};$  ▷ conviction decays with switch risk
26     $\sigma \leftarrow d_k \cdot c;$ 
27   $\Sigma.\text{append}(\sigma);$ 
28  ▷ Step 4 -- Compound portfolio over next window
29  for each  $r_t \in r^{(\text{pf})}[e : e + w]$  do
30     $V.\text{append}(V[-1] \times (1 + \sigma r_t));$ 
31 return  $V, \Sigma$ 

```

4.2 Ledoit-Wolf test on sharpe ratios

Conducted Ledoit-Wolf test to compare the Sharpe ratios of the different trading strategies and see if the differences in performance are statistically significant. Conducted test with a confidence level of 95% and reported the p-values for each pairwise comparison between the trading strategies and the long-only benchmark. Used hac method as well as the studentized circular bootstrap approach.

4.3 FX Data

I have tested different trading strategies based on regime detection using the sWk-means clustering algorithm with UnifOrtho projections and the implied probability strategy derived from the SW distance to the centroids and compared it to a long-only benchmark on the training period from 2006 to 2020 (FX data).

4.3.1 Equity Data

I have tested different trading strategies based on regime detection using the sWk-means clustering algorithm with UnifOrtho projections and the implied probability strategy derived from the SW distance to the centroids and compared it to a long-only benchmark on the training period from 2006 to 2020 (3483 observations for the American Equity Data).

American Indices Equity No significant statistically difference in sharpe ratios between the trading strategies and a long-only benchmark.

4.4 Tuning the hyperparameters for the trading strategies

Tested multiple values for the hyperparameters of the trading strategies, such as the entry and hold

4.4.1 Tuning the weights for ensemble methods

Tested multiple combination of weights for the ensemble methods, which combines the binary regime long/short strategy and the implied probability strategy with different variations (hysteresis, continuous, conviction). The weights are ordered as following :

$[w_{binary}, w_{implied_probability(hysteresis)}, w_{implied_probability(continuous)}, w_{implied_probability(conviction)}]$. We have tested the following weights:

- $[0.25, 0.25, 0.25, 0.25]$,
- $[0.30, 0.10, 0.30, 0.30]$,
- $[0.20, 0.30, 0.20, 0.30]$,
- $[0.30, 0.20, 0.30, 0.20]$,
- $[0.10, 0.40, 0.10, 0.40]$

4.4.2 Tuning the gradient weight for implied probability methods

Tested multiple values for the gradient weight α in the implied probability strategy with gradient. The values tested are the following

$$\alpha \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\} \quad (12)$$

Optimal value for α is determined based on the highest Sharpe ratio achieved during the backtesting period. For the American equity data we have determined that the optimal value is $\alpha = 0.5$. As most of the variants of implied probability strategies (continus and hysteresis), have a peak performance at $\alpha = 0.5$.

$\alpha = 0.0$ corresponds to a pure bayesian approach. It captures current cluster membership, regime persistence and more generally where the market is. $\alpha = 1.0$ corresponds to a pure gradient approach. It captures the direction of the market (getting closer or further from a certain regime) and therefore captures the momentum of the market.

Hence an optimal value of $\alpha = 0.5$ suggests that regime identification and regime momentum are equally predictive. We could also thinkg that $\mathbb{P}_{Bayers}$ and $\mathbb{P}_{Grad}$ are not redundant and capture different information about the market.

4.4.3 Tuning the temperature parameter τ and τ_{grad} for the implied probability strategy

Performed grid search for τ taken as adaptative to the switch probability as well as for τ_{grad} taken as adaptative to the gradient values. The optimal values for τ and τ_{grad} are determined based on the highest Sharpe ratio achieved during the backtesting period. For the American equity data, we have determined that the optimal value for τ is 0.01 and the optimal value for τ_{grad} is 0.01.

5 Experiment 5: Interpretability Analysis of Sliced Wasserstein Regime Detection Algorithm

We label the regimes using weekly equity data (S&P 500, NASDAQ, Dow Jones) on the period 2006 to 2020. We further consider the following macro feature set:

- VIX Index
- 10-year Treasury Yield
- 5-year Treasury Yield
- 30-year Treasury Yield
- Fed Funds Rate
- Manufacturing PMI
- Manufacturing Employment
- DXY Dollar Index
- Gold Spot Price
- Brent Crude Oil Spot Price

After having regrouped the data from the following macro-indicators from the appropriate time period, we fit the feature set to the regime labels using the following models:

- OLS Linear Regression (probabilistic target (Probability of being in bear regime))
- Logistic Regression (binary target (Bull/Bear))
- Random Forest Classifier (binary target (Bull/Bear))

In particular, we are analysing the feature importance of the different macro-indicators during the bear regimes using the SHAP analysis toolbox. We observe that in particular during predicted bear regimes by the sliced-Wasserstein k-means clustering algorithm, the VIX index is a particularly important feature in the regime classification for all three models.

6 Appendix

References

- [1] Patric Bonnier, Patrick Kidger, Imanol Perez Arribas, Cristopher Salvi, and Terry Lyons. Deep signature transforms. *arXiv preprint arXiv:1905.08494*, 2019.
- [2] B. Horvath, A. Muguruza, and M. Tomas. Wasserstein k-means for clustering distributions. *Journal of Machine Learning Research*, 2024.
- [3] O. Ledoit and M. Wolf. Robust performance hypothesis testing with the sharpe ratio. *Journal of Empirical Finance*, 15(5):850–859, 2008.
- [4] Qimeng Luan and James Hamp. Automated regime classification in multidimensional time series data using sliced wasserstein k-means clustering. *Data Science in Finance and Economics*, 5(3):387–418, 2025.
- [5] Rémi Bardenet Vladimir Petrovic and Agnès Desolneux. Repulsive monte carlo on the sphere for the sliced wasserstein distance. *arXiv preprint arXiv:2509.10166*, 2025.